

DENVER STAPLETON, CO, USA

WMO: 724690

Lat: 39.767N

Lon: 104.869W

Elev: 1612

StdP: 83.40

Time Zone: -7.00 (NAM)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-18.6	-14.9	-22.0	0.6	-14.8	-19.1	0.8	-10.1	12.2	4.9	10.6	4.9	2.2	180	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	34.4	15.9	32.9	15.6	31.4	15.3	18.0	27.7	17.4	27.0	16.8	26.5	4.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	13.5	19.5	14.7	12.7	19.4	13.8	12.0	19.7	57.3	27.8	55.0	26.8	53.1	26.7	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	8.8	7.7	-23.6	37.6	4.6	1.4	-26.8	38.6	-29.5	39.4	-32.1	40.2	-35.4	41.3		
			-23.9	19.8	4.5	1.1	-27.1	20.6	-29.7	21.3	-32.2	21.9	-35.5	22.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	0.2	1.3	5.7	10.0	15.1	20.7	22.8	21.9	17.4	10.9	3.8	-0.9	(6)
	DBStd	9.75	5.84	6.77	5.31	5.21	4.12	3.67	2.81	2.72	4.09	5.16	5.87	6.44	(7)
	HDD10.0	1359	303	245	149	63	9	0	0	0	3	50	198	338	(8)
	HDD18.3	3148	561	476	391	251	114	19	4	6	63	231	437	596	(9)
	CDD10.0	1651	1	2	16	64	166	320	396	370	226	78	11	1	(10)
	CDD18.3	401	0	0	0	2	13	89	142	118	36	1	0	0	(11)
	CDH23.3	5106	0	0	4	77	269	1148	1679	1273	549	104	3	0	(12)
	CDH26.7	2111	0	0	0	11	61	512	804	532	175	17	0	0	(13)
(14) Wind	WSAvg	3.6	3.6	3.4	3.9	4.0	4.1	3.8	3.6	3.4	3.3	3.2	3.4	3.2	(14)
(15) Precipitation	PrecAvg	386	12	15	28	47	57	40	48	43	37	27	18	13	(15)
	PrecMax	546	29	49	98	148	144	87	104	102	197	63	59	68	(16)
	PrecMin	205	0	0	0	0	10	0	1	3	0	2	5	0	(17)
	PrecStd	82	8	11	25	30	32	26	29	26	37	17	14	13	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.6	19.9	24.1	28.0	30.7	36.5	36.8	35.0	32.7	28.5	23.2	17.9	(19)
		MCWB	6.0	7.6	8.9	11.6	12.9	15.5	16.3	16.2	15.5	11.5	8.5	6.3	(20)
	2%	DB	14.7	16.7	21.4	25.7	28.3	33.9	34.7	33.2	30.6	26.3	19.5	14.7	(21)
		MCWB	4.8	6.1	7.9	10.8	12.5	15.1	16.1	15.8	14.3	11.0	7.9	5.0	(22)
	5%	DB	12.2	14.1	18.7	23.0	26.5	31.9	33.2	31.7	28.7	23.9	16.4	11.4	(23)
		MCWB	3.4	4.9	6.7	10.1	12.5	15.0	15.9	15.7	13.7	10.2	6.4	3.5	(24)
	10%	DB	9.7	11.6	15.7	20.4	24.3	29.7	31.4	29.9	26.8	21.1	13.2	8.4	(25)
		MCWB	2.4	3.7	5.7	8.8	12.0	14.9	15.8	15.5	13.3	9.2	4.9	1.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	8.1	9.6	12.6	15.0	18.3	19.1	19.1	17.4	12.9	10.1	6.9	(27)
		MCDB	16.5	18.9	22.8	24.8	24.1	28.1	30.0	27.1	29.4	22.9	19.5	16.3	(28)
	2%	WB	4.9	6.3	8.1	11.4	14.1	17.4	18.1	18.0	15.8	11.7	8.4	5.2	(29)
		MCDB	14.0	15.8	19.8	23.5	23.3	27.7	28.1	26.7	25.6	23.2	17.9	13.9	(30)
	5%	WB	3.6	5.1	7.1	10.4	13.4	16.6	17.5	17.3	15.0	10.8	6.9	3.7	(31)
		MCDB	11.8	13.7	17.6	21.6	23.1	26.7	27.3	26.1	24.8	21.8	15.7	11.1	(32)
	10%	WB	2.4	3.8	6.0	9.4	12.7	15.8	16.8	16.7	14.1	9.9	5.3	1.8	(33)
		MCDB	9.2	11.2	15.1	19.5	22.3	26.1	27.0	25.7	24.1	20.4	12.4	8.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.6	12.8	13.6	13.9	14.1	15.1	15.5	14.7	15.5	15.3	13.3	13.5	(35)
		MCDBR	16.5	16.8	18.4	17.7	17.5	18.2	18.2	17.4	18.1	19.7	17.7	17.2	(36)
	5% WB	MCWBR	9.2	8.8	8.6	7.2	6.4	5.6	5.3	5.2	6.3	8.3	9.0	9.5	(37)
		MCWBR	16.0	16.3	17.4	16.5	15.2	15.6	14.4	15.5	17.9	17.0	16.9	16.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.241	0.249	0.271	0.302	0.320	0.328	0.358	0.348	0.310	0.273	0.254	0.243	(40)
		taud	2.621	2.607	2.551	2.477	2.487	2.478	2.437	2.452	2.538	2.626	2.604	2.609	(41)
	Ebn at Noon		963	996	996	975	956	944	912	915	938	951	935	934	(42)
		Edh at Noon	66	77	92	106	107	109	112	107	92	75	67	62	(43)
(44) All-Sky Solar Radiation	RadAvg		2.58	3.57	4.78	5.66	6.25	7.14	6.76	6.05	5.24	3.86	2.77	2.25	(44)
	RadStd		0.11	0.19	0.36	0.39	0.49	0.34	0.30	0.21	0.28	0.24	0.16	0.13	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+58	N/A

Nomenclature: See separate page